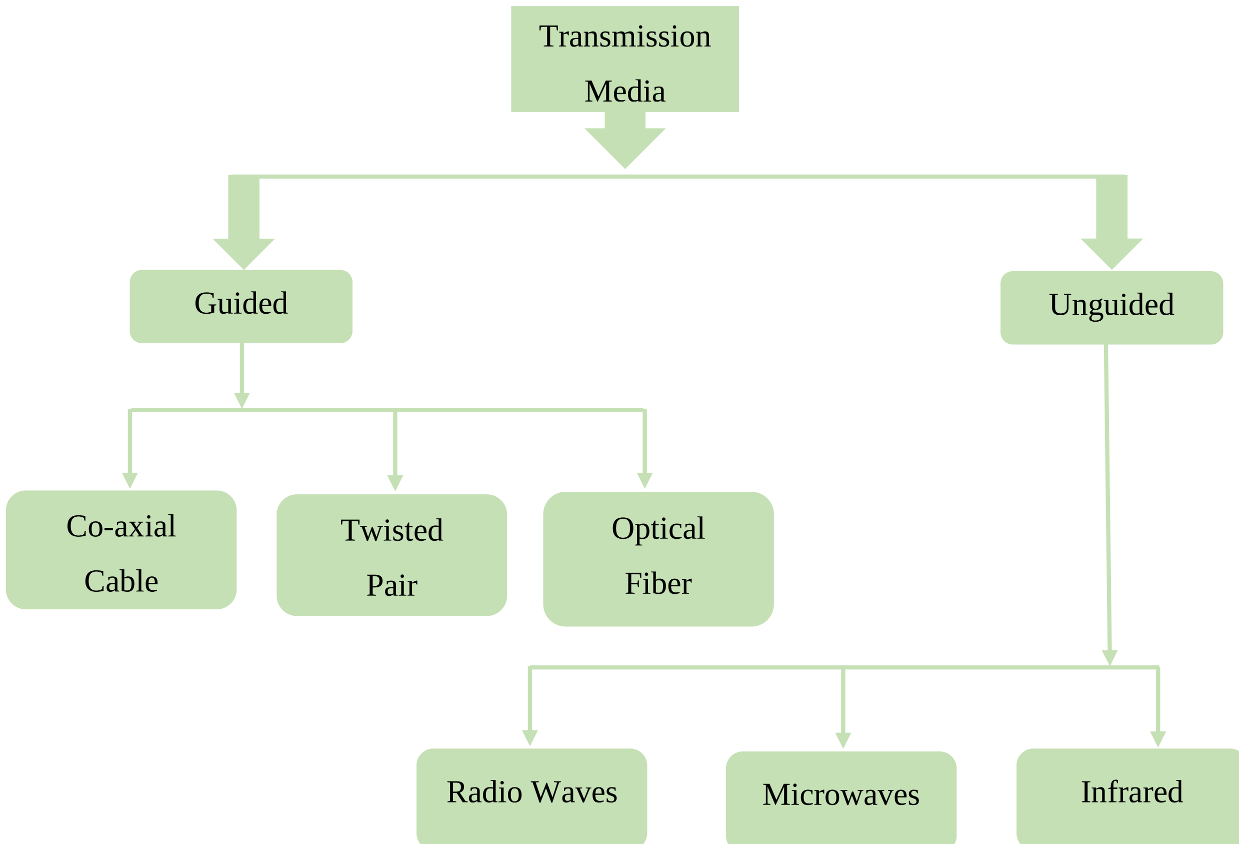
Chapter 3: Transmission Media and Data Compression

A transmission medium is the physical pathway that connects the transmitter and the receiver, serving as the channel through which data is transmitted from one location to another.



3.1Guided Transmission Media: Co-axial cable, Twisted pair, Optical fiber

Guided Transmission Media refers to the physical mediums that guide signals along a specific path, usually through wires or cables.

Features of guided transmission media are:

- High Speed
- Secure
- Used for comparatively shorter distance

Types of Guided Transmission Media

1. **Co-axial Cable**
2. **Twisted Pair**
3. **Optical Fiber**

Co-axial Cable

A coaxial cable is a type of guided transmission medium that consists of a central copper wire surrounded by insulation, a metallic shield, and an outer plastic cover. The central copper conductor carries the signals, while the insulating layer surrounds it to protect and isolate it. Around this insulator, a braided metal shield is placed to block external electrical interference and reduce crosstalk between cables.

Structure of Co-axial Cable:

- Copper Conductor: This is the main core made of copper, responsible for carrying and transmitting the data signals.
- Insulator: A dielectric plastic layer that surrounds the copper conductor, designed to keep proper spacing between the central conductor and the outer shield.
- Braided Mesh: A copper braid that surrounds the insulator, providing protection against electromagnetic interference. It helps block external EMI and also prevents internal signals from leaking out of the coaxial cable.
- Protective Plastic Layer: The outermost polymer coating that serves to safeguard the inner components of the cable from physical damage and environmental wear.

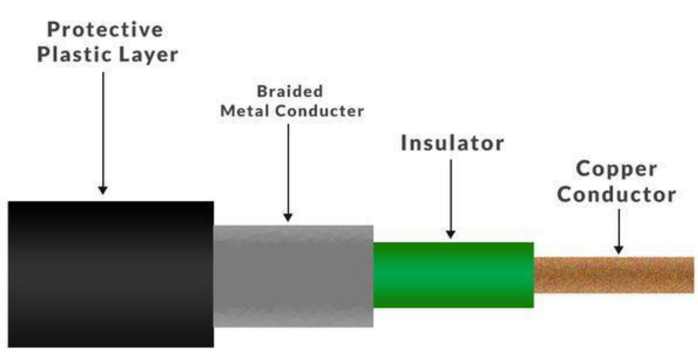


Figure: Co-axial Cable

Uses of Coaxial Cable

1. **Internet:** Coaxial cables are commonly used to transmit internet signals, with RG-6 cables being the standard type for this purpose.
2. **Television:** Coaxial cable used for television would be 75 Ohm and RG-6 coaxial cable.
3. **Long-distance Telephone Transmission:** Coaxial cable has traditionally been an important part of the long-distance telephone network. A coaxial cable can carry over 10,000 voice channels simultaneously.

Advantages of Coaxial cable

- Coaxial cables can handle high bandwidth.
- They are easy to install.
- Coaxial cables are more reliable and durable due to their strong resistance to physical damage.
- They are less vulnerable to noise, crosstalk, and electromagnetic interference.
- Coaxial cables can carry multiple channels simultaneously.

Twisted Pair Cable

It consists of 2 separately insulated conductor wires twisted about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. One of the conductors is used to carry the signal and the other is used as a ground reference only.

Figure: Twisted Pair Cable

- The purpose of twisting the wires is to reduce electrical interference (or noise) from similar pairs of close by.
- The most common application of twisted pair is the telephone system.
- Twisted pairs can run several kilometers without amplification, but for longer distances repeaters are needed.
- The twisted pair cable can be broadly categorized into the following two types:
  I. Unshielded Twisted Pair (UTP) Cable
  II. Shielded Twisted Pair (STP) Cable

I. Unshielded Twisted Pair (UTP) Cable

- UTP is the most common type of twisted pair cable used in telecommunication networks.
- The UTP consists of two copper conductors, each having their own insulating material, intertwined with each other to cancel induced currents.
- The reason for placing twist in the pair of wires is to minimize the vulnerability of the twisted pair cable to external electrical noise.
- Its frequency range is suitable for data transmission as well as voice transmission.

Advantages:

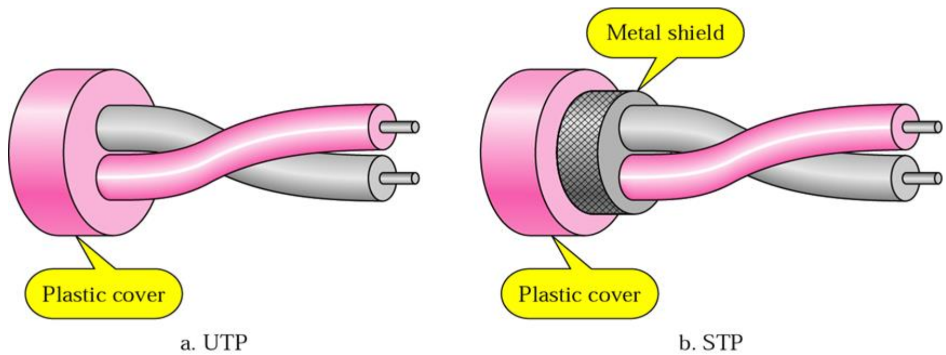
- Low cost
- It is easy to install.
- It is basically used in LAN implementation.

The Electronic Industries Association (EIA) has developed five standards or categories for UTP:

- Category-1: This category is used for voice transmission and it is basically used in telephone system. It is efficient only for low speed data transmission.
- Category-2: This category is used for voice transmission, but it is equally preferred for data transmission of upto 4mbps.
- Category-3: This category is used for data transmission because it provides high bandwidth to achieve high data transfer rates. It is standard cable for most telephone systems.
- Category-4: It provides much higher data transfer rate of 16mbps.
- Category-5: It provides a maximum data transfer rate of 100mbps. For UTP generally RJ45 connectors are used.

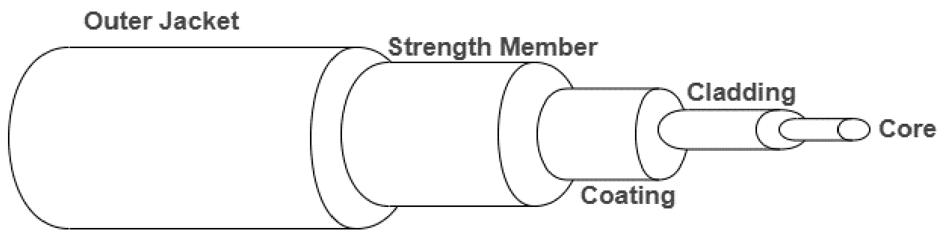
II. Shielded Twisted Pair (STP) Cable

- In STP, each pair of insulated conductors is encased in a metal shield to prevent crosstalk.
- The quality aspect of STP is same as the UTP but the shield must be connected to ground.
- STP cables use same connectors as UTP, and are more expensive than UTP cable.
- It is used for both analog and digital transmission.
- It is said to be a balanced transmission medium.



Optical Fiber

An optical fiber cable operates based on the principle of total internal reflection of light within a glass core. The core is enclosed by a cladding layer made of glass or plastic with a lower refractive index. Optical fiber cables are widely used for high-capacity data transmission and can support either unidirectional or bidirectional communication. Wavelength Division Multiplexing (WDM) technology enables both unidirectional and bidirectional modes of data transmission over the fiber.



OPTICAL FIBER CABLE

Advantages of Optical Fiber

- Enormous potential bandwidth
- Small size and weight
- Electrical isolation
- Immunity to interference and crosstalk
- Signal security
- Low transmission loss
- Ruggedness and flexibility
- System reliability and ease of maintenance
- Potential low cost
- Faster data transmission
- Fire and explosion safety
- Allows to transmit information over long distances
- Taping is more easily detected.

Disadvantages of Optical Fiber

- More complex transmitter and reception equipment are essential.
- Splicing is more difficult and complicated.
- More expensive for short distance
- A small bend on fiber cause radiation loss.

3.2 Unguided Transmission Media: Radio waves, Microwaves, Infrared

It is also known as wireless or unguided transmission media, where electromagnetic signals are transmitted without the need for any physical medium.

Features

- The signal is broadcasted through air
- Less Secure
- Used for larger distance

Radio Waves

Radio waves are the most extensively used type of unguided transmission media due to their ability to cover long distances and penetrate solid structures like buildings.

They are widely utilized in applications such as AM/FM radio broadcasting, television transmission, and mobile communication systems.

Types of Radio waves:

- Short wave: AM Radio
- Very High Frequency (VHF): FM Radio/TV
- Ultra High Frequency (UHF): TV

Advantages

- Can be generated easily and travel long distances
- Ideal for broadcasting over wide areas
- Can penetrate buildings and other obstacles

Microwaves

Microwave communication uses high-frequency radio waves for point-to-point communication.
Microwaves are widely used in mobile communication systems, satellite networks, and data transmission between buildings or long distances.
Microwave communication systems require a clear line of sight between the transmitting and receiving antennas, making them vulnerable to interference from physical objects like mountains or buildings.
They offer high-speed data transmission and are widely used for long-distance telephone communication

Advantages

- Supports high-speed data transmission
- Suitable for long-distance communication
- Can handle a large volume of data traffic

Infrared

Infrared (IR) communication is a short-range wireless technology commonly used in remote controls, wireless keyboards, and other small personal devices. Infrared signals travel in straight lines and cannot penetrate walls or other solid objects, which limits their range but also makes them more secure from external interference.

Advantages

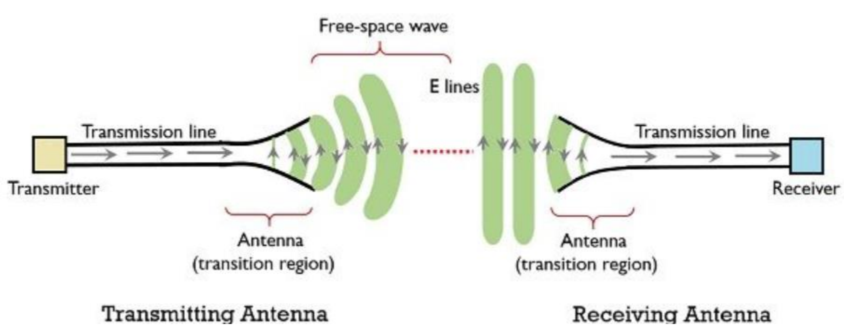
- Immune to interference from other radio signals
- Provides a secure connection for short-range communication

3.3Antenna Basics, Satellite Communication, Bluetooth, Wi-Fi

Antenna Basics

An antenna is an electrical device which converts electric power into radio waves and vice-versa. An antenna is a:

- Transducer
- Metallic device
- Gateway of radio link
- Medium between guided wave and electromagnetic wave
- Main component of wireless communication



Need for Antenna

Antennas are considered to be the basic building block of a wireless communication system. As a signal in free space propagates in the form of an electromagnetic wave then it requires some transmitting as well as receiving ends. So, an antenna allows the propagation of electromagnetic waves from one end to another, in the form of EMW without the need for a wiring system. It acts as a transducer that converts RF signal into EM wave at the transmitter and EM wave back to an electrical signal at the receiver.

Functions of Antenna

- Transducer: It acts as a transducer as it converts one form of energy into a different form.

  At the transmitting end, an electrical signal is transformed into a radio wave, and the receiving end changes radio waves back to electrical form.

- Radiator and Sensor: The transmitting and receiving antennas act as radiators and sensors of electromagnetic waves, respectively. The transmitting antenna radiates the electromagnetic wave in free space, while the receiving antenna senses the presence of an electromagnetic wave in the space and collects it.
- Coupler: Another important feature possessed by the antennas is coupling. It acts as a coupler between the device that generates the RF signal and free space or free space and transmission lines.

Basic Antenna Parameters

- Radiation Pattern: It is a mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates.

- Radiation Power Density: It is a quantity that is used to describe the power associated with an electromagnetic wave.
- Directivity: It is defined as the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions.
- Gain: In a given direction, gain of an antenna is defined as the ratio of the intensity in a given direction to the ratio of the radiation intensity that would be obtained if the power accepted by the antenna were radiated isotropically.
- Antenna Efficiency: It is defined as the ratio of the aperture effective area to its actual physical area.

Satellite Communication

A satellite is generally defined as a smaller object that orbits a larger body in space. For instance, the Moon is a natural satellite of the Earth.

Communication refers to the exchange of information between two or more entities through a specific medium or channel. It involves the processes of transmitting, receiving, and interpreting information. When communication occurs between two Earth stations via a satellite, it is referred to as satellite communication. This type of communication utilizes electromagnetic waves as carrier signals to transmit information such as voice, audio, video, or data between the Earth and space, and vice versa.

Need of Satellite Communication

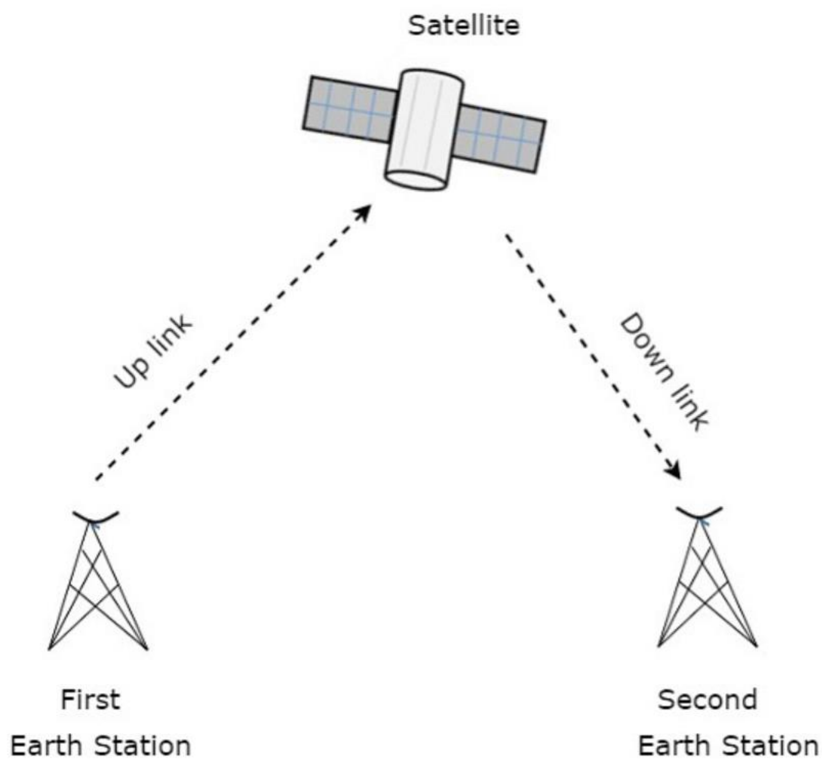
Traditionally, two primary modes of signal propagation have been used for communication over limited distances: Ground wave and Sky wave propagation.

However, both these methods have a maximum range limitation of around 1500 kilometers due to the Earth's curvature and signal degradation. Satellite communication addresses and overcomes these limitations by enabling long-distance communication well beyond the line of sight. Because satellites are positioned at considerable altitudes above the Earth's surface, they facilitate seamless communication between widely separated Earth stations. This capability effectively eliminates the distance and curvature constraints associated with ground-based propagation methods.

Working

A satellite is an object that orbits another body along a defined trajectory. A communication satellite specifically functions as a microwave repeater station positioned in space. It plays a vital role in supporting telecommunications, radio and television broadcasting, as well as internet services.

A repeater is an electronic circuit designed to amplify a received signal before retransmitting it. In the context of satellite communication, this repeater functions as a transponder, a device that not only amplifies the incoming signal but also shifts its frequency band before retransmission to avoid interference. The frequency used to transmit a signal from an Earth station to the satellite is known as the uplink frequency**,** whereas the frequency at which the satellite sends the signal back to Earth is referred to as the downlink frequency.



Satellite communication starts at an Earth station, which is a facility equipped to transmit and receive signals to and from a satellite in orbit around the Earth. These stations transmit information to the satellite using high-frequency, high-power signals typically in the gigahertz (GHz) range. The satellite receives these signals, processes them, and then retransmits them back to Earth. The signals are received by other Earth stations located within the satellite's coverage area. This coverage area, known as the **satellite's footprint**, defines the region on the Earth's surface where the satellite signal is strong enough to be effectively received.

Advantages of Satellite Communication

- Area of coverage is more than that of terrestrial systems
- Each and every corner of the earth can be covered
- Transmission cost is independent of coverage area
- More bandwidth and broadcasting possibilities.

Disadvantages of Satellite Communication

- Launching of satellites into orbits is a costly process.
- Propagation delay of satellite systems is more than that of conventional terrestrial systems.
- Difficult to provide repairing activities if any problem occurs in a satellite system.
- Free space loss is more

Applications of Satellite Communication

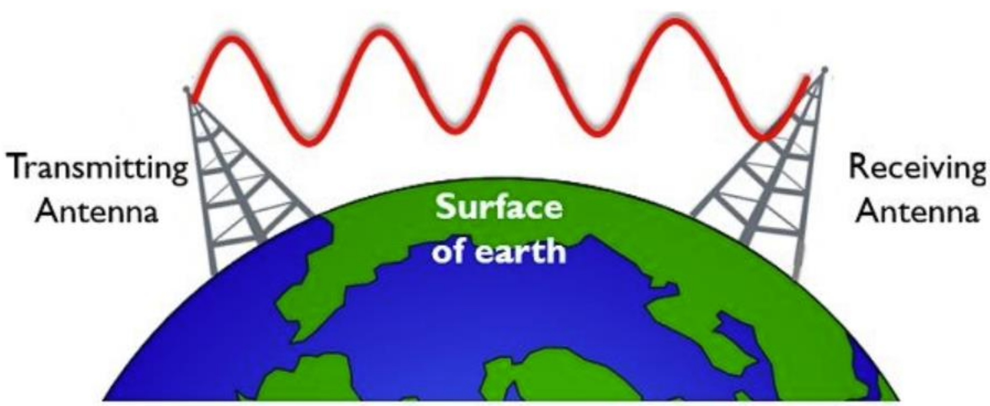
- Radio broadcasting and voice communications
- TV broadcasting such as Direct To Home (DTH)
- Internet applications such as providing Internet connection for data transfer, GPS applications, Internet surfing, etc.
- Military applications and navigations
- Remote sensing applications
- Weather condition monitoring & Forecasting

Wireless Propagation

1. Ground Wave Propagation

Ground Wave Propagation refers to a type of radio wave transmission that occurs in the region between the Earth's surface and the ionosphere. In simpler terms, this mode of propagation involves electromagnetic waves traveling close to the Earth's surface. It is also commonly referred to as surface wave propagation.

Ground wave propagation is generally a low-frequency signal propagation technique that shows suitability with the small range of operation in relation to distance. The operating frequency range in the case of ground wave propagation usually lies in the range from kHz to a few MHz (generally up to 2 MHz).



<u>Basic Properties</u>

- Main Antenna: Marconi, Whip
- Frequency Band: Very Low Frequency (VLF), Low Frequency (LF), Medium Frequency (MF).
- Frequency: Upto 2MHz
- Polarization: Vertical
- Transmission Mode: Amplitude Modulation
- Utilized for short range communication
- Signal attenuation due to absorption and diffraction

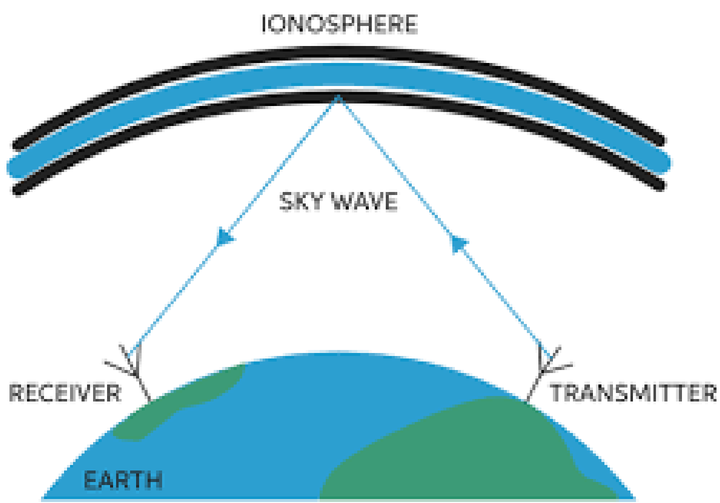
- Frequency < 15 MHz; Less attenuation
- Frequency > 2MHz; not good for ground wave

Factors Affecting Ground Wave

- Nature of Ground: ground conductivity
- Polarization
- Weather
- Frequency

2. Sky Wave Propagation

A type of radio wave communication in which the electromagnetic wave propagates due to the reflection mechanism of the ionospheric layer of the atmosphere is known as sky wave propagation. Due to propagation through the ionosphere, it is also known as ionospheric wave propagation.

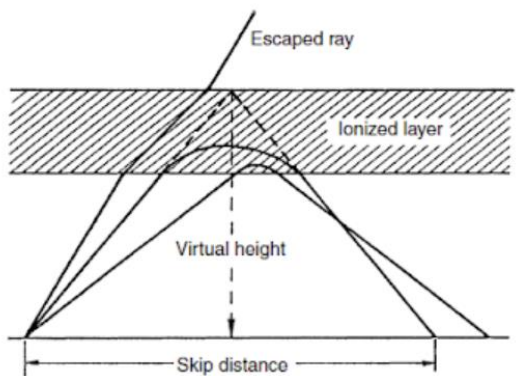


We know that the ionosphere is present in the upper atmospheric region and is composed of ionized layers. Generally, the ionosphere consists of 4 different layers namely D, E, F_1, and F_2. These layers are present at different heights from the surface of the earth.

Basically the ionosphere is said to be extended from 60 to 400 Km from the surface of the earth.

Each layer has a different concentration of atoms in a way that the ionized layer which is present nearer to the surface of the earth has the highest number of neutral atoms. While the middle layer has moderate concentration and the outermost layer consists of a very less number of neutral atoms. Sky wave propagation is suitable for the frequency range from 3MHz to 30 MHz. But for signal frequency greater than 30 MHz, space wave propagation is used.

An important factor of ionospheric wave propagation is skip distance. And skip distance is defined as the minimum distance on the surface of the earth from where the signal is transmitted and the reflected signal from the ionosphere has been received.



Frequency Bands

Band Name	Symbols	Frequency range
Very low frequency	VLF	3 to 30 kHz
Low frequency	LF	30 to 300 kHz
Medium frequency	MF	300 to 3000 kHz
High frequency	HF	3 to 30 MHz
Very high frequency	VHF	30 to 300 MHz
Ultra high frequency	UHF	300 to 3000 MHz
Super high frequency	SHF	3 to 30 GHz
Extremely high frequency	EHF	30 to 300 GHz
Terahertz (ITU, 2015b)	THz	300 to 3000 GHz

Source: ITU (2012), Radio Regulations Article 2